Unit: 6 Unit 4: Absolute Value **Name:** ________________
Topic: Absolute Value Function and its Graph **Date:** ___________

Advanced (Habanero)

Q1. What is the vertex of the graph of $y = |x - 2|$?
(A) (0, 2)
(B) (2, 0)
(C) (0, 0)
(D) (2, 2)
(E) (-2, 0)

Q2. Which of the following is the graph of $y = |x| + 1$?
(A) A vertical shift of the graph of $y = |x|$ down 1 unit
(B) A vertical shift of the graph of $y = |x|$ up 1 unit
(C) A horizontal shift of the graph of $y = |x|$ left 1 unit
(D) A horizontal shift of the graph of $y = |x|$ right 1 unit
(E) A reflection of the graph of $y = |x|$ across the x-axis

Q3. What is the equation of the axis of symmetry of the graph of $y = |x - 3|$?
(A) $x = -3$
(B) $x = 0$
(C) $x = 3$
(D) $x = 6$
(E) $x = -6$

Q4. Which of the following functions has a graph that opens downward?
(A) $y = |x|$
(B) $y = -|x|$
(C) $y = |x - 2|$
(D) $y = |x| + 1$
(E) $y = -|x| + 1$

Q5. What is the equation of the graph of $y = |x|$ after a reflection across the y-axis?
(A) $y = |x|$
(B) $y = |-x|$
(C) $y = |x| + 1$
(D) $y = -|x|$
(E) $y = |x - 1|$

Q6. Which of the following is a transformation of the graph of $y = |x|$?
(A) $y = |x| + 2$
(B) $y = |x| - 2$
(C) $y = |x - 2|$
(D) $y = 2|x|$
(E) All of the above

Q7. What is the vertex of the graph of $y = |x + 2| - 1$?
(A) $(-2, 0)$
(B) $(-2, -1)$
(C) $(-2, 1)$
(D) $(0, -1)$
(E) $(2, -1)$

Q8. Which of the following equations represents a horizontal shift of the graph of $y = |x|$?
(A) $y = |x| + 1$
(B) $y = |x| - 1$
(C) $y = |x - 1|$
(D) $y = |x + 1|$
(E) $y = 2|x|$

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 2| + 1$ and identify the vertex, axis of symmetry, and direction of opening.

Q10. Describe the transformations applied to the graph of $y = |x|$ to obtain the graph of $y = 2|x - 1| + 3$.

Chef's Tips

- Emphasize the importance of vertex, axis of symmetry, and direction of opening in absolute value functions.
- Provide examples of real-world applications of absolute value functions.
- Encourage students to use graphing calculators to visualize and explore absolute value functions.